Building your own tokamak from scratch

A beginner's guide to HOT-Jass web

Seven numbers describe a tokamak well enough to estimate how hot its plasma gets, how much current it can drive and whether it can make fusion power. This guide starts from those seven numbers, builds a machine that does not exist yet – **MY-ST**, a compact spherical tokamak – from an empty screen, and shows how HOT-Jass tells you when an input or a result needs a second look.

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Contents

1 Seven numbers make a tokamak **2**

1.1 What follows from the seven numbers 2

2 The empty rail: NEW MACHINE **3**

3 A worked example: MY-ST **4**

3.1 Step 1 – type the seven numbers 5

3.2 Step 2 – add a beam (and aim it) 5

3.3 Step 3 – Start, and read the results 5

3.4 Step 4 – an experiment: halve the density 7

4 Making it realistic **8**

5 The "Worth checking" flags **9**

5.1 Before Start 10

5.2 After the run 10

5.3 PONTE: the demanding profile 10

6 Common surprises **10**

7 Where to go next **11**

1 Seven numbers make a tokamak

A tokamak looks complicated, but for a first estimate of its performance you need only **seven numbers**: three for its shape, two for its magnetic field, one for the plasma it holds, and one for how well it keeps heat in.

	Parameter	What it sets	MY-ST
R_0	major radius	size; volume; how far the beam travels	0.9 m
a	minor radius	size; aspect ratio $A = R_0/a$	0.55 m
κ	elongation	volume and current capacity (a taller plasma holds more)	2.0
B_0	toroidal field	confinement; pressure limit (β)	1.0 T
I_p	plasma current	confinement; stability (q); density limit	1.2 MA
n_e	electron density	how many particles share the heat; beam stopping	$6 \times 10^{19}\,\mathrm{m}^{-3}$ (centre)
τ_E	energy confinement time	how long the heat stays in	40 ms

The eighth ingredient, the **heating power** P, is not a property of the machine: it is what you *do* to it. In HOT-Jass that is mostly neutral-beam injection (NBI), with optional ECRH and ICRH.

1.1 What follows from the seven numbers

Everything below is a back-of-the-envelope formula, evaluated for MY-ST. The code solves the same physics more carefully; where it matters, HOT-Jass's own result for the same machine (Section 3) is quoted alongside.

Plasma volume.

$$V \simeq 2\pi^2 R_0\, a^2 \kappa = 2\pi^2 \times 0.9 \times 0.55^2 \times 2.0 = 10.75\ \mathrm{m}^3.$$

Safety factor (stability). The field lines wind around the torus; the edge safety factor q counts how many times they go round the long way for one turn the short way:

$$q_{\mathrm{cyl}} \simeq \frac{5\, a^2 B_0}{R_0\, I_p[\mathrm{MA}]} \frac{1+\kappa^2}{2} = \frac{5 \times 0.55^2 \times 1.0}{0.9 \times 1.2} \times \frac{1+4}{2} = 3.50.$$

Below $q \approx 2$ the plasma is kink-unstable and disrupts; real machines keep $q_{95} \gtrsim 3$. A spherical tokamak gets a bonus from its tight aspect ratio: HOT-Jass's shaped formula gives $q_{95} = 6.9$ for MY-ST.

Density limit. The Greenwald density caps the line-averaged density:

$$n_{\mathrm{GW}}\,[10^{20}\,\mathrm{m}^{-3}] = \frac{I_p[\mathrm{MA}]}{\pi a^2} = \frac{1.2}{\pi \times 0.55^2} = 1.26.$$

MY-ST's line average is $3.2 \times 10^{19}\,\mathrm{m}^{-3}$, a comfortable $n/n_{\mathrm{GW}} = 0.25$.

Temperature: the energy balance. In steady state the heating replaces what leaks out. With $n_i \simeq n_e = n$ and $T_i \simeq T_e = T$, the stored energy is $W = 3\langle n\rangle\langle T\rangle V$, so

$$W = P\,\tau_E \quad\Longrightarrow\quad \langle T\rangle \simeq \frac{P\,\tau_E}{3\,\langle n\rangle V}.$$

A central density $n_{e0} = 6\times10^{19}$ with the default profile $n(\rho) = n_{e0}(1-\rho^2)^{2p_n}$, $p_n = 1$, averages to $\langle n\rangle = n_{e0}/(1+2p_n) = 2\times10^{19}\,\mathrm{m}^{-3}$. With $P = 3$ MW and $\tau_E = 40$ ms:

$$\langle T\rangle \simeq \frac{3\times10^{6}\times0.04}{3\times2\times10^{19}\times10.75\times1.602\times10^{-16}\,\mathrm{J/keV}} \approx 1.2\ \mathrm{keV}.$$

HOT-Jass gives $\langle T_e\rangle = 1.29$ and $\langle T_i\rangle = 1.13$ keV. The centre is about three times hotter than the average: $T_{e0} = 3.9$ keV.

Pressure limit. The plasma pressure is measured against the magnetic pressure, $\beta_t = 2\mu_0\langle p\rangle/B_0^2$, and normalised to the current:

$$\beta_N = \beta_t[\%]\,\frac{a\,B_0}{I_p[\mathrm{MA}]}.$$

Ideal-MHD stability needs $\beta_N \lesssim 3.5$ in a conventional tokamak and somewhat more ($\sim$ 5–6) in a spherical one. With its fast beam ions MY-ST reaches $\beta_t = 5.1\%$, $\beta_N = 5.1\times0.55/1.2 = 2.3$.

Fusion power and gain. For a deuterium–tritium plasma,

$$P_{\mathrm{fus}} = n_D\,n_T\,\langle\sigma v\rangle(T)\,E_{\mathrm{fus}}\,V, \qquad E_{\mathrm{fus}} = 17.6\ \mathrm{MeV}, \qquad Q = P_{\mathrm{fus}}/P_{\mathrm{heat}}.$$

$\langle\sigma v\rangle$ rises steeply up to $T\sim20$ keV, so the temperature – that is, $P\tau_E/n$ – decides almost everything. In a beam-heated plasma the fast beam ions also fuse with the thermal ions (*beam-target* fusion), which at a few keV is far more than the thermal plasma makes on its own.

Where does τ_E come from?

τ_E is the one number you cannot read off a drawing of the machine: it depends on turbulence in the plasma. You can either *type it in* (the simplest start, used below) or let HOT-Jass compute it from a *scaling law* fitted to many experiments, e.g. for spherical tokamaks (Kaye 2006, NSTX H-mode)

$$\tau_E = 4.69\times10^{-9}\,I_p[\mathrm{A}]^{0.57}\,B_0^{1.08}\,n_e[\mathrm{m}^{-3}]^{0.44}\,P[\mathrm{W}]^{-0.73}\ \mathrm{s},$$

or for conventional tokamaks the ITER scaling IPB98(y,2)

$$\tau_E = 0.0562\,I_p^{0.93}B_0^{0.15}n_{19}^{0.41}P^{-0.69}R_0^{1.97}\kappa^{0.78}\varepsilon^{0.58}M^{0.19}\ \mathrm{s} \quad (I_p\ \mathrm{MA},\ P\ \mathrm{MW},\ \varepsilon = a/R_0).$$

HOT-Jass keeps separate electron and ion confinement times, so its rail has two cells, **tauE,e** and **tauE,i**. In this guide we type the **same** τ_E into both.

2 The empty rail: NEW MACHINE

HOT-Jass opens in the **NEW MACHINE** state (the magenta button above the presets, Fig. 1): every parameter in the **INPUT DATA** rail on the left is empty and every model is set to its simplest option. Clicking **NEW MACHINE** at any time brings you back here; the eight buttons below it load preset devices instead (seven existing machines and the PONTE design study).

The colour of each value cell tells you what HOT-Jass needs:

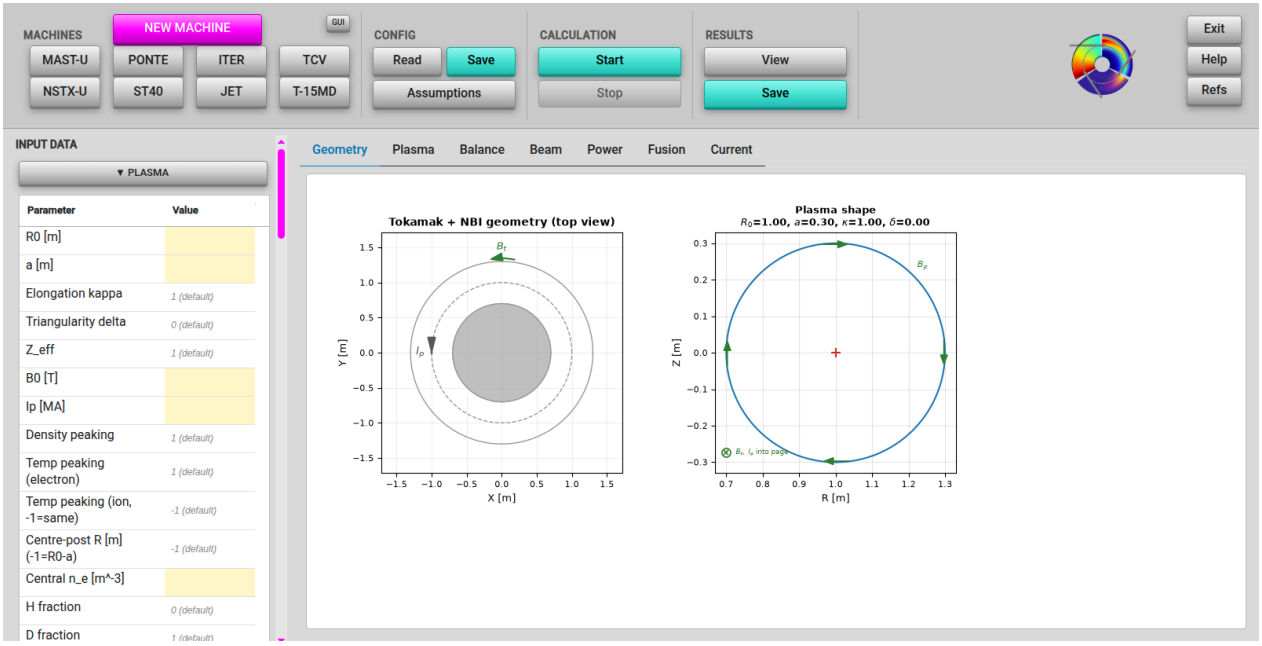


Figure 1: HOT-Jass at start-up: NEW MACHINE selected, the rail empty. Yellow cells must be filled in; gray italic text is the default an empty cell will use.

Cell	Meaning
	Required and still empty. Start is blocked until you fill it in.
1 (def.)	**Optional**: if you leave it empty, the calculation uses the gray value shown.
not used	**Not used** with the current settings (a beam that is switched off, a model you have not selected).
abc	**Invalid**: not a number, out of range, or fractions that do not add up. Blocks Start.
1.5	**Worth checking**: allowed, but probably not what you meant (Section 5). Never blocks Start.
0.9	Changed by hand since the last preset or file was loaded.

Only a handful of cells are always required: R_0, a, B_0, I_p and the central density n_{e0} – plus at least one heating power above zero. A few more become required when you choose something that needs them: the two τ_E cells while **Confinement** is **Fixed tauE (input)**, and the beam energy and tangency radius of a beam whose power is above zero. An empty power counts as zero, that is, *switched off*.

Tip

Values change colour as soon as you change something they depend on: type a beam power and that beam's cells turn from gray *not used* to yellow (required) or gray-italic defaults.

3 A worked example: MY-ST

MY-ST is not one of the presets: a compact spherical tokamak ($A = 1.64$) of about MAST size, heated by a single 3 MW deuterium beam.

3.1 Step 1 – type the seven numbers

Open the **PLASMA** section and type:

cell	**R0**	**a**	**kappa**	**B0**	**Ip**	**Central n_e**	**tauE,e**	**tauE,i**
value	0.9	0.55	2.0	1.0	1.2	6e19	0.04	0.04

Click a cell, type, press Enter or click the next cell. Press **Start**. HOT-Jass refuses, and the **Input check** dialog says why (Fig. 2): there is **no heating**. It also has a first piece of advice – more on that in Step 2.

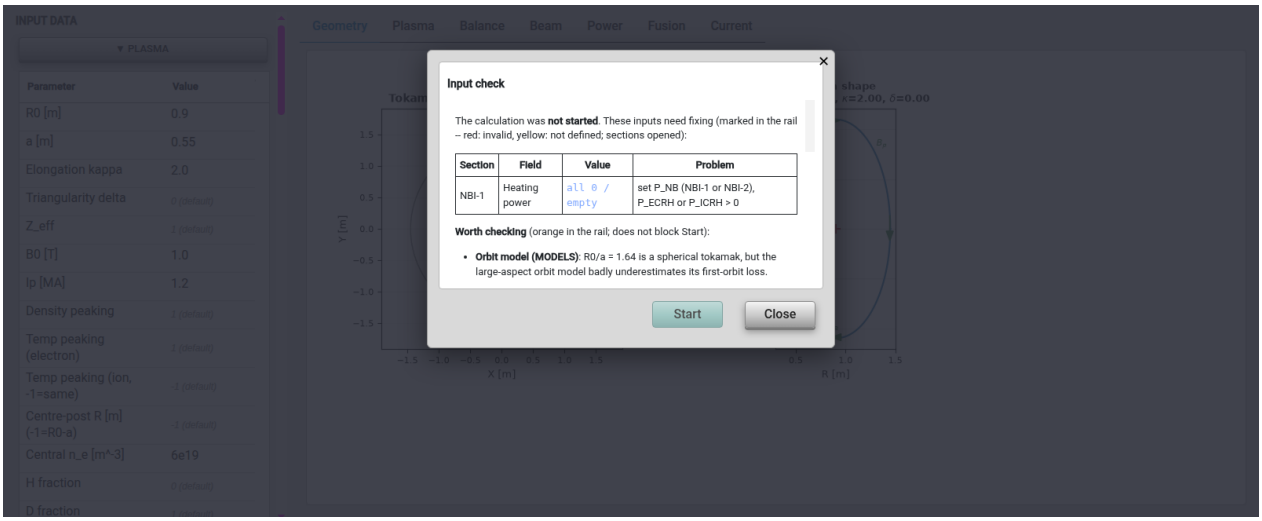


Figure 2: Step 1: seven numbers, but no heating. The Input check lists what blocks Start (top) and what is merely worth checking (bottom).

3.2 Step 2 – add a beam (and aim it)

Open **NBI-1** and type **P_NB** = 3 MW. The beam is now on, and two of its cells turn yellow: **E_b**, its energy – type 60 keV – and **tangent R**, the radius at which the beam line passes closest to the torus axis. The others show their defaults (a deuterium beam, all at full energy). Let us make a typical beginner's mistake and type $R_{\mathrm{tan}} = 1.5$ m. The cell turns red (Fig. 3): the plasma ends at $R_0 + a = 1.45$ m, so this beam would *miss it*, and HOT-Jass will not start until you correct it. The Geometry tab shows the same: the beam line passes outside the plasma.

Correct it to $R_{\mathrm{tan}} = 0.9$ m (aimed at the magnetic axis, a common choice). Then open **MODELS** and change two of the simplest-option models:

- **Orbit model** → **ST orbits – P_phi orbit boundary**. The Input check already warned (Fig. 2): with $R_0/a < 2$ the large-aspect model badly underestimates how many fast ions are lost on their first orbit.
- **Shine-through** → **Suzuki**. The **Manual** model, with its default fraction 0, does not trace the beam at all: it counts the whole beam as absorbed, wherever it goes. Suzuki calculates how much of the beam passes straight through.

3.3 Step 3 – Start, and read the results

Press **Start**. After a few seconds the result tabs fill in.

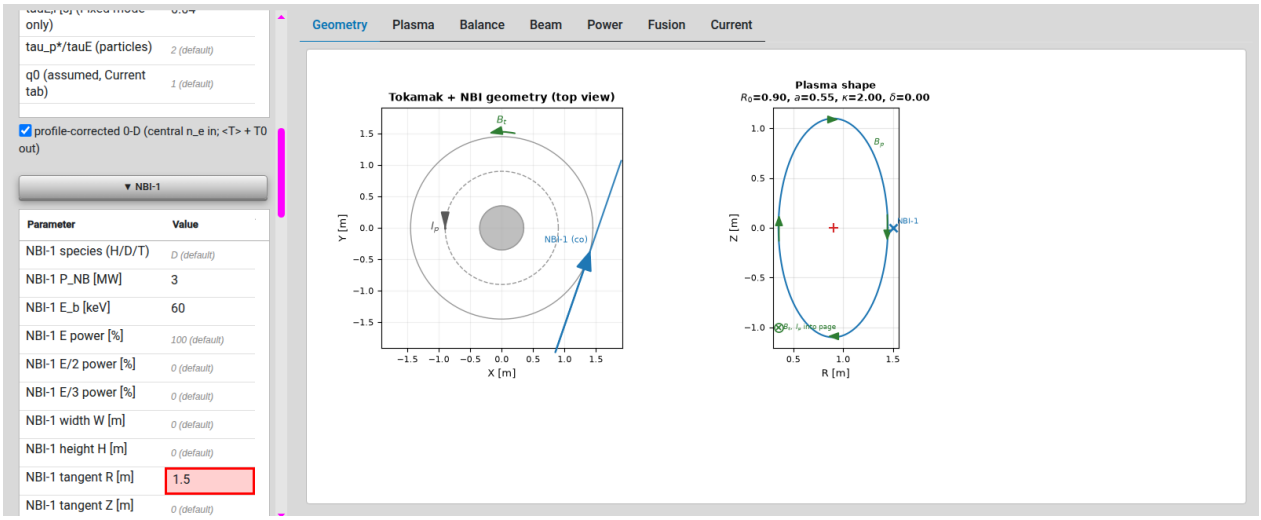


Figure 3: Step 2: a beam aimed at $R_{\tan} = 1.5$ m, outside the plasma. The cell is red – Start is blocked – and the Geometry view shows the miss.

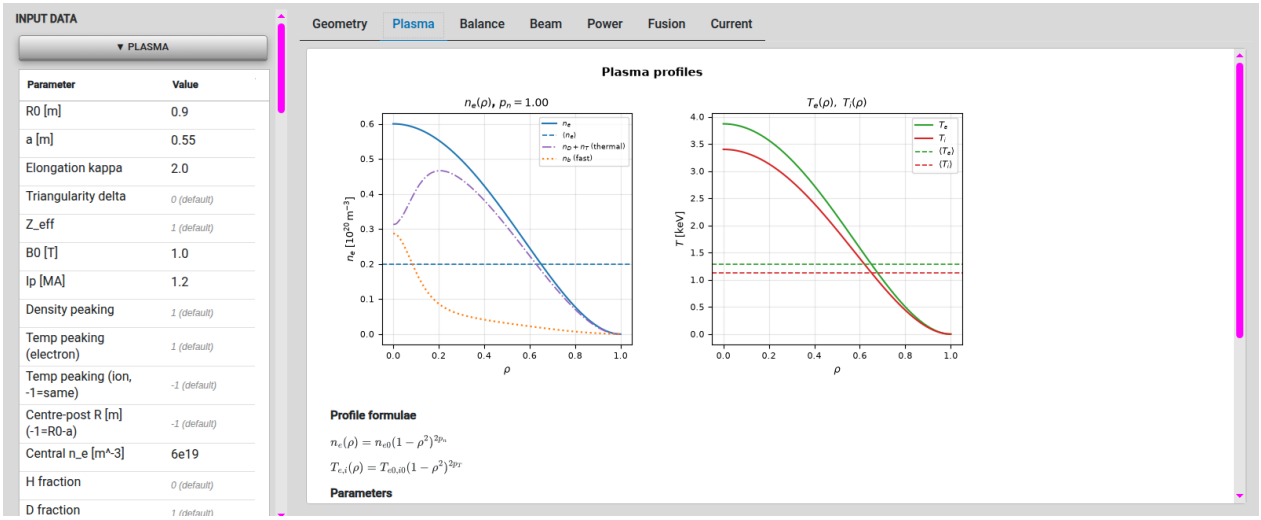


Figure 4: Step 3, Plasma tab: density and temperature profiles of MY-ST. Dashed lines are the volume averages.

Plasma (Fig. 4): $T_{e0} = 3.87$ keV, $T_{i0} = 3.40$ keV in the centre; the averages $\langle T_e \rangle = 1.29$, $\langle T_i \rangle = 1.13$ keV agree with the 1.2 keV of the simple estimate in Section 1. The orange dotted curve is the density of fast beam ions.

Power: of the 3 MW injected, only 1.2% shines through and 0.2% is lost on first orbits; 1.67 MW heats the electrons and 1.29 MW the ions.

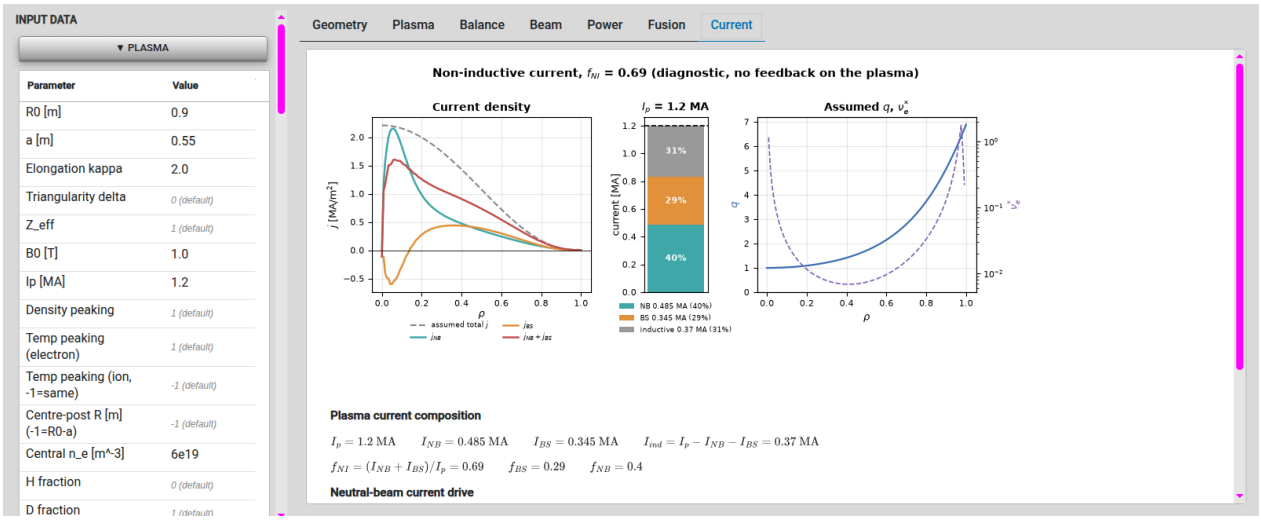


Figure 5: Step 3, Current tab: the beam drives 0.49 MA and the bootstrap effect 0.35 MA of the 1.2 MA plasma current.

Current (Fig. 5): the beam drives $I_{NB} = 0.485$ MA and the pressure gradient another $I_{BS} = 0.345$ MA (bootstrap current), so 69% of the plasma current needs no transformer.

A first flag. The Beam tab carries an orange note: *fast ions are 12% of all ions.* The model treats the beam ions as a small population slowing down in the plasma; above $\sim 10\%$ that assumption starts to weaken. At 12% it is a reminder, not an alarm.

3.4 Step 4 – an experiment: halve the density

Now change **Central n_e** to 3×10^{19} and press **Start** again.

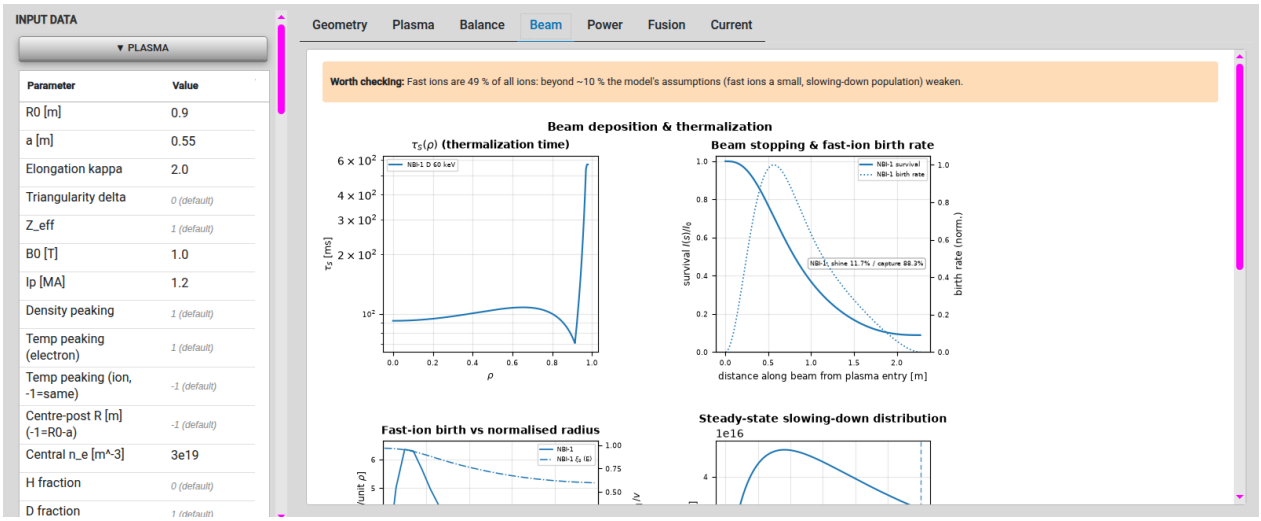


Figure 6: Step 4, Beam tab at half the density: 11.7% of the beam shines through, and fast ions are now half of all ions.

Everything moves at once (Fig. 6):

- Fewer particles stop the beam: shine-through rises from 1.2% to 11.7%.
- The same beam power now heats far fewer thermal ions: T_{i0} jumps from 3.4 to 13.1 keV.
- Fast ions become 49% of all ions – the flag now says the model is being stretched, and the result is a rough upper estimate.
- The beam drives 3.7 MA in a 1.2 MA plasma ($f_{NI} = 3.2$): the plasma is *overdriven*, and the assumed current profile is no longer self-consistent.

The lesson: NBI needs enough density to stop the beam and to share its energy. Put n_{e0} back to 6×10^{19}.

4 Making it realistic

The run in Step 3 is a valid first estimate. To make it closer to a real plasma, replace the simplest choices one at a time, and watch what each one does.

Confinement from a scaling law. In **MODELS**, set **Confinement** to **Kaye NSTX H-mode** (the spherical tokamak scaling of Section 1); the two τ_E cells turn gray, *not used*. Tick **electron-ion equipartition**, the heat exchange between electrons and ions. The scaling gives MY-ST $\tau_E = 97$ ms rather than our guessed 40 ms.

Shape and profiles. Give the plasma some triangularity, **delta** = 0.4, and a flatter density profile, **Density peaking** = 0.5. Keep **profile-corrected 0-D** ticked (the default for NEW MACHINE): without it the power balance assumes flat profiles and the peaking changes little.

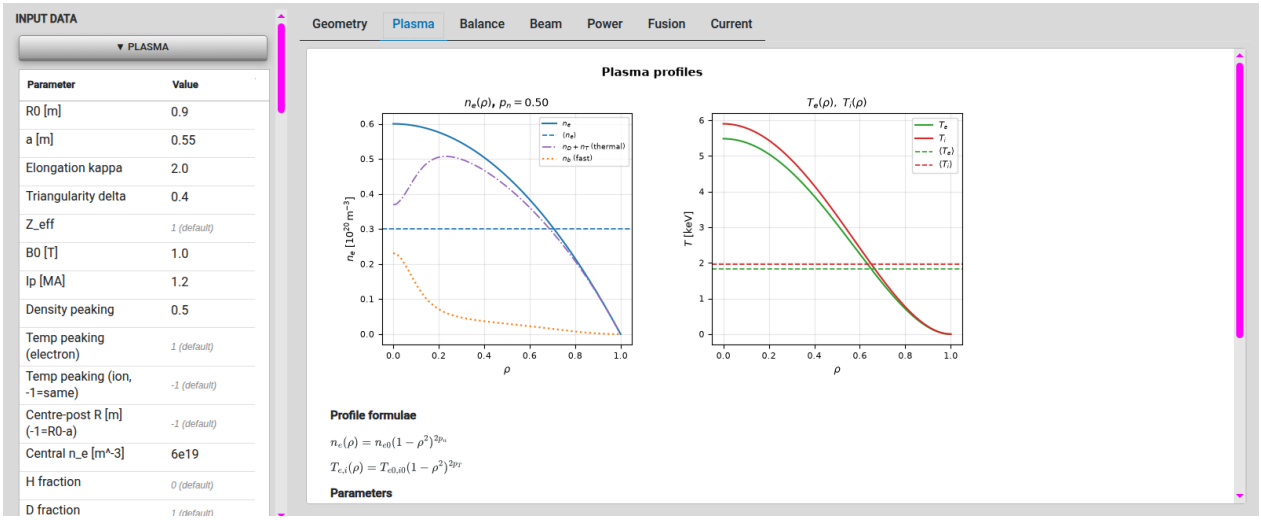


Figure 7: MY-ST with Kaye NSTX H-mode confinement, equipartition, $\delta = 0.4$ and density peaking 0.5.

Result (Fig. 7): $T_{e0} = 5.5$ keV, $T_{i0} = 5.9$ keV, $\beta_N = 3.8$, $f_{NI} = 0.84$, fast ions 7% – and no flags.

Fuel. So far MY-ST burns deuterium only (the default fuel mix is pure D) and makes a mere 2 kW of D-D fusion. Set the fuel to **D fraction** = 0.5, **T fraction** = 0.5:

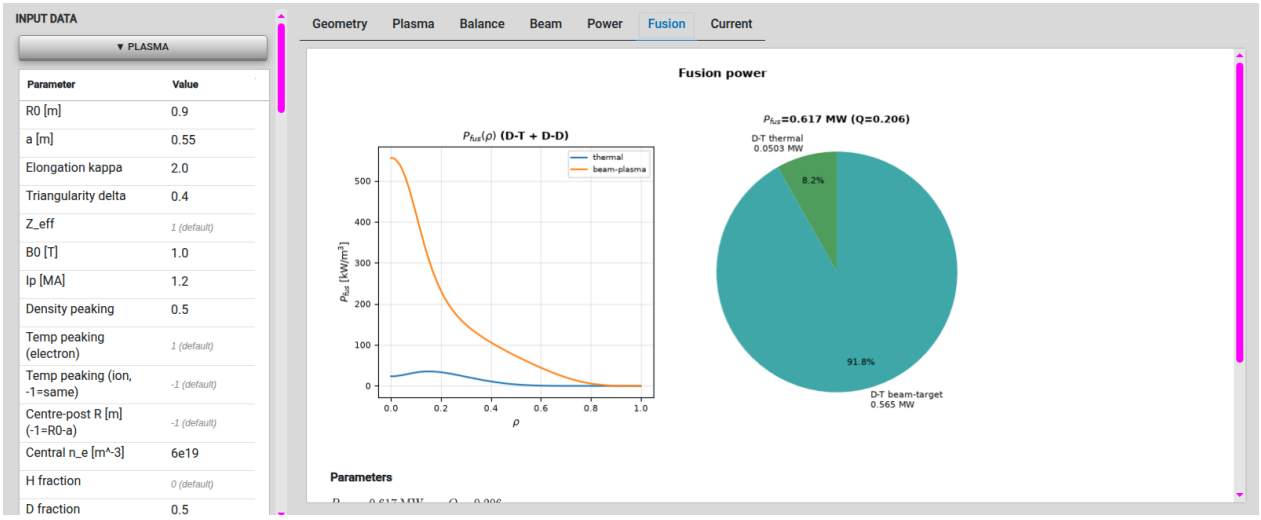


Figure 8: MY-ST with a 50:50 D-T plasma: 0.62 MW of fusion power, 92% of it from beam ions fusing with the thermal tritium.

$P_{\mathrm{fus}} = 0.62$ MW, $Q = 0.21$ (Fig. 8). 92% of it is *beam-target* fusion: at $T \approx 6$ keV the thermal plasma is too cold to fuse much on its own, but the 60 keV deuterons are not. This is why beam-driven spherical tokamaks are attractive as compact neutron sources.

Other cells worth filling in.

- **Z_eff** (default 1, a perfectly clean plasma): real plasmas have $Z_{\mathrm{eff}} \approx 1.5$–2.
- **E, E/2, E/3 power**: a positive-ion beam source also delivers atoms at half and a third of the full energy, e.g. 80/15/5 %.
- **width W**, **height H**: the beam's cross-section (default: a single ray).
- **Centre-post R**: the radius of the central column, if it is not $R_0 - a$.

Save your machine

Config → **Save** writes all inputs, including the empty cells, to a JSON file; **Read** loads it back. The file names the device "NEW MACHINE"; change its `device_name` field to, say, "MY-ST" and that name is shown when you load it.

5 The "Worth checking" flags

HOT-Jass checks your inputs before every Start, and the results after every run, for the most common beginner traps. Flags never block a run; they colour the cell orange, appear as a notification, and are listed in the **Input check** dialog or at the top of the result tab concerned.

5.1 Before Start

Flag	When	What to do
Kink limit	$q_{95} < 2$	Lower I_p, or raise B_0, a or κ.
Greenwald	$\langle n_e \rangle_{\text{line}} > n_{\text{GW}}$	Lower n_{e0} or raise I_p.
Beam hits the post	$R_{\text{tan}} <$ centre-post radius	Aim it further out.
Orbit model	$R_0/a < 2$ with the large-aspect model	Choose an ST orbit model.
Peaking has little effect	peaking set, profile correction off	Tick **profile-corrected 0-D**.
q_0 has no effect	$q_0 \geq q_{\text{cyl}}$	Use $q_0 < q_{\text{cyl}}$ (Section 6).
q_0 below 1	$q_0 < 1$	Sawteeth keep $q_0 \approx 1$ in a real plasma; use $q_0 = 1$.

Two impossible inputs are not flags but errors (red), and Start is blocked until you correct them: $a \geq R_0$, and a beam aimed outside the plasma, $R_{\text{tan}} > R_0 + a$.

5.2 After the run

Flag	When	Tab	Meaning
Shine-through	$> 25\%$ of P_{NB}	Power	Beam too energetic for the density; the wall takes the power.
First-orbit loss	$> 30\%$ of P_{NB}	Power	I_p too low to hold the fast ions.
β_N	> 3.5 (> 5.5 if $R_0/a < 2$)	Plasma	Ideal-MHD unstable.
Fast ions	$> 10\%$ of all ions	Beam	The slowing-down model is stretched.

5.3 PONTE: the demanding profile

The **PONTE** preset is a design study that pushes several limits at once. Its users get *more* flags, marked [**PONTE strict**]: tighter limits ($q_{95} < 3$, $n/n_{\text{GW}} > 0.8$, shine-through $> 10\%$, first-orbit loss $> 15\%$, fast ions $> 5\%$, $\beta_N > 3.0$ or 4.5) and extra checks: the choice of models (fixed τ_E, equipartition off, an orbit model other than P_ϕ, D and T beams without beam-beam fusion), a strongly hot-ion plasma ($T_{i0}/T_{e0} > 2$), $T_{i0} > 30$ keV, thermal ions below 70%, fast-ion energy above the thermal one, an overdriven current ($f_{NI} > 1$), and more than 0.5 MW of shine-through. Seeing several warnings on PONTE is expected: they show where that design is bold.

6 Common surprises

"I changed q_0 and nothing happened."
q_0 is the on-axis safety factor of an *assumed* current profile $j \propto (1 - \rho^2)^\nu$ with $\nu = q_{\text{cyl}}/q_0 - 1$. It only affects the bootstrap current (Current tab) and, for the P_ϕ orbit model or tilted beams, the fast-ion pitch. Once $q_0 \geq q_{\text{cyl}}$, $\nu = 0$ – the profile is flat – and larger q_0 changes nothing. An empty κ (i.e. 1) makes q_{cyl} small, so this happens easily.

"Density peaking does nothing."
With **profile-corrected 0-D** off, the power balance uses flat profiles; the peaking then only shapes the deposition and current profiles.

"Shine-through is zero, whatever I do."
That is the **Manual** shine-through model: it uses the fixed fraction you type (default 0) and never traces the beam. Choose Suzuki, Janev or Riviere.

"Why are some empty cells gray, not yellow?"
They have a default (shown in gray italics) or are not used by the current settings. Every default the calculation actually used is listed in the Input check dialog and in a notification at Start.

"The Current tab says $f_{NI} > 1$."
The beam and bootstrap currents exceed I_p. The calculation does not feed the current back into the plasma, so this is a sign to lower the beam power or raise I_p, not a solution.

7 Where to go next

- Load the presets – MAST-U, NSTX-U and ST40 are spherical tokamaks like MY-ST; compare their numbers with yours.
- **Config** → **Assumptions** lists every physics assumption and formula the calculation uses.
- **Refs** links the papers behind the models and the validation data.
- **Results** → **Save** exports a PDF report, U-files and an NBI_OUT file for transport codes such as ASTRA.